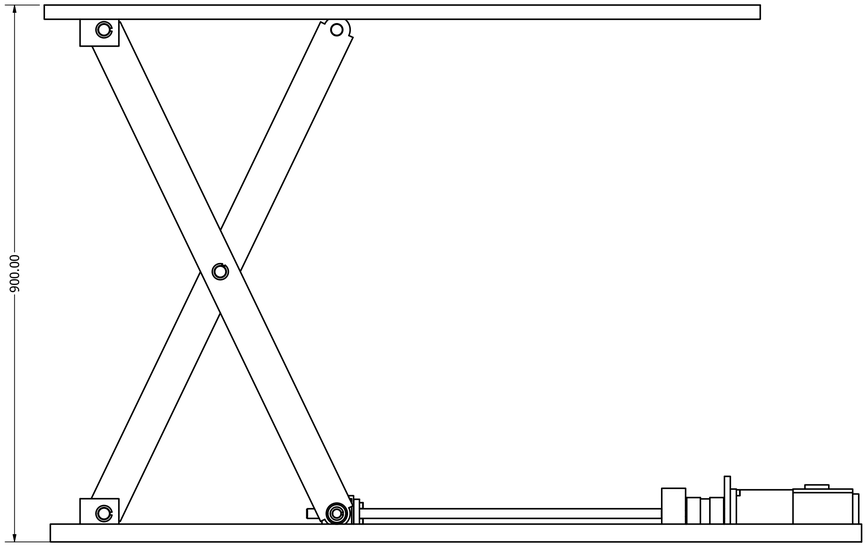
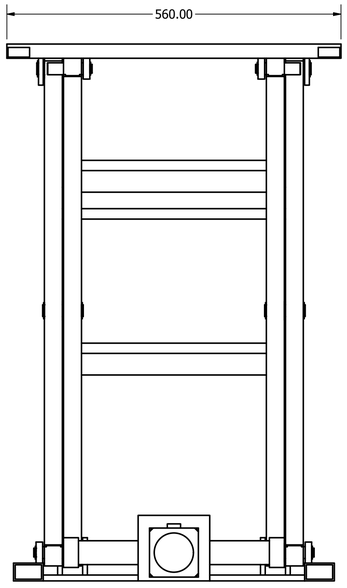


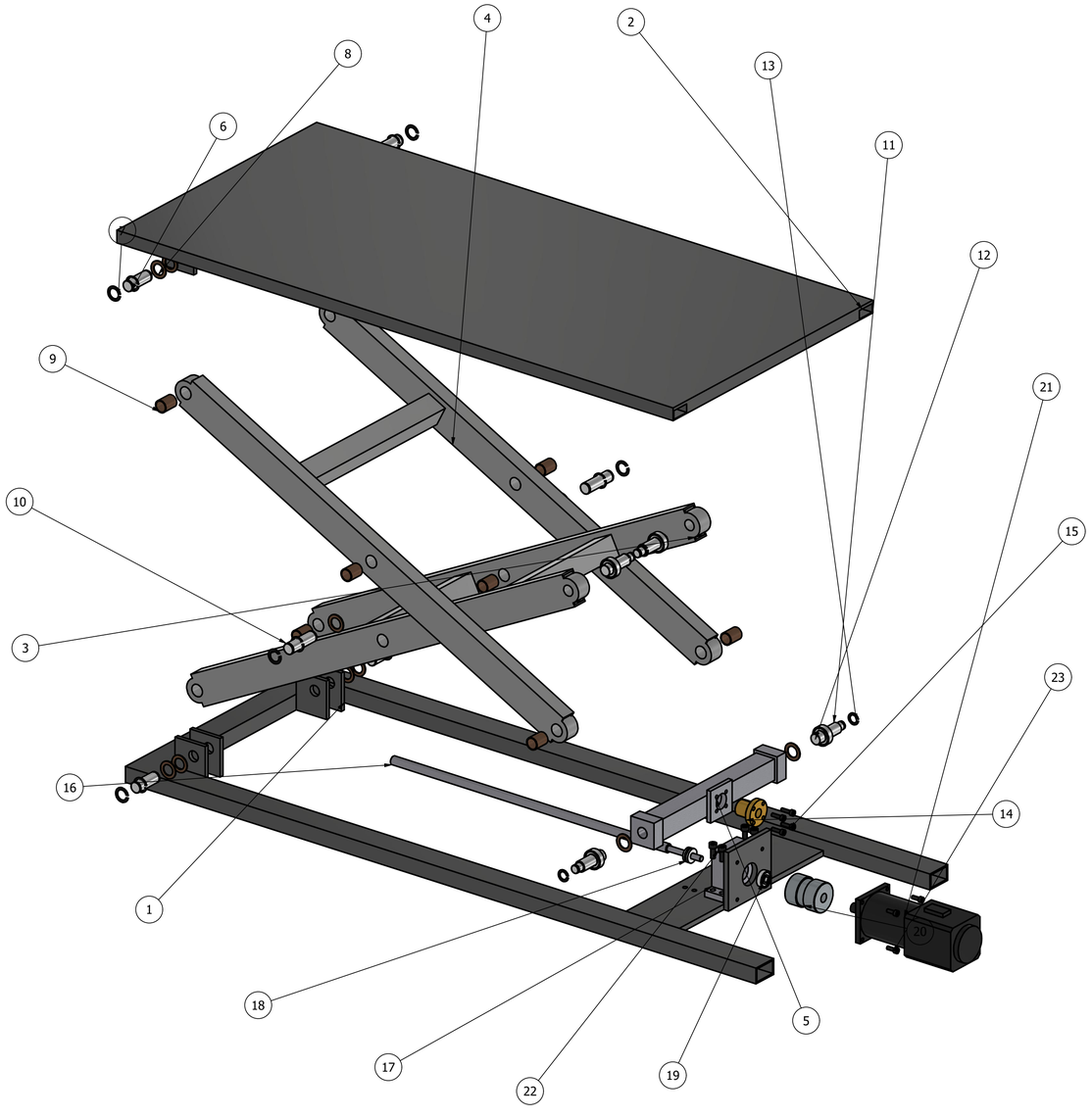






SCISSORLIFT - ELECTRIC BENCH SCISSOR LIFT TABLE, 150 KG RATED (DESIGNED AND CALCULATED, NOT BUILT)
DECK 1200 x 560, HEIGHT 250 - 900 MM; ONE SCISSOR STAGE, ARMS RHS 60x30x3 S355, 900 PIN CENTRES, ARM ANGLE 10.3 - 64.3 DEG
DRIVE: TR16x4 LEAD SCREW (COMPRESSION) PUSHING THE NUT CARRIAGE, 51200 + 6200 IN THE DRIVE BLOCK, L-095 COUPLING,
STEPPERONLINE 24 V 200 W BLDC + 71.5:1 PLANETARY (33 NM RATED, 42 RPM); LIFT TIME 177 S; DRIVER TORQUE LIMIT 21 NM
SCREW FORCE 10.7 KN AT 250 MM, 0.91 KN AT 900 MM; PEAK SCREW TORQUE 15.0 NM (MU 0.10) / 19.0 NM (MU 0.15) (CALC.)
ARMS: PEAK 71 MPA (SF 5.0 ON YIELD), PINS SF >= 3.1, TIP-OVER SF 3.0 WITH A 10 % PUSH AT 900 MM (CALC.)
PIVOTS: 20 MM 42CrMo4 PINS IN SKF PCM 202330 E BUSHINGS, PCMW THRUST WASHERS; ROLLERS 6202-2RS ON 15 MM PINS.
ASSEMBLY PARAMETER: Lift (DECK HEIGHT); theta = asin((Lift - 89) / 900). DO NOT WORK UNDER THE DECK WITHOUT THE MAINTENANCE PROP.

DRAWN Jay Patel	10/29/2026		
CHECKED		TITLE	
QA		ScissorLift - general assembly	
MFG			
APPROVED		SIZE D	DWG NO SL-DWG-001 REV
		SCALE 1 / 5	SHEET 1 OF 2



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	SL-101	Base frame, welded: 2 x RHS 50x30x3 rails 1360 long (roller tracks), RHS front member, 10 mm drive plate + motor plate, 4 x 8 mm clevis plates for the inner-arm pivot (20 H8)
2	1	SL-102	Platform, welded: 1200 x 560 x 4 deck plate, 2 x RHS 40x20x2 roller rails, 2 edge tubes, 2 end members, 4 x 8 mm clevis plates for the outer-arm pivot (20 H8)
3	1	SL-103	Inner arm frame, welded: 2 x RHS 60x30x3 S355 arms (900 pin centres), 45 OD bosses (23 H7 bushing seats at the pivot and centre, 20 P7 for the roller pins), SHS 40x40x3 cross tubes at 360 / 700
4	1	SL-104	Outer arm frame, welded: 2 x RHS 60x30x3 S355 arms (900 pin centres), 45 OD bosses with 23 H7 bushing seats, SHS 40x40x3 cross tube at 250 from the top pivot
5	1	SL-105	Nut carriage, welded: SHS 40x40x3 cross tube, 2 x 50x50x25 C45 end blocks (20 P7 roller-pin bores), 8 mm nut plate (4 x M6 on 38 PCD); pivot axis = screw axis, so the screw force makes no moment on the nut
6	4	SL-301	Pivot pin 20 f7 x 56, 42CrMo4 QT ground: clevis pivots (base / inner arm and platform / outer arm), 2 x DIN 471-20
7	12	SL-501	Retaining ring DIN 471 20 x 1.2 (groove 19.0 x 1.3)
8	12	SL-405	Thrust washer 20 x 32 x 1.5, composite (SKF PCMW class), one each side of every arm boss
9	10	SL-404	Composite plain bushing SKF PCM 202330 E 20 x 23 x 30 (PTFE / bronze / steel back), arm pivots; p static 250 MPa, dynamic 80 MPa, pv 1.6 MPa m/s
10	2	SL-302	Centre pin 20 f7 x 68, 42CrMo4 QT ground: inner / outer arm crossing, 2 x DIN 471-20
11	4	SL-303	Roller pin, 42CrMo4 QT: 20 p6 x 60 pressed into the carriage block / inner-arm boss, 15 k6 seat for the 6202-2RS track roller, DIN 471-15 groove
12	4	SL-401	Deep groove ball bearing 6202-2RS 15 x 35 x 11 as track roller (C 8.06 kN, C0 3.75 kN); INA LR202-2RSR crowned track roller is the drop-in upgrade
13	4	SL-502	Retaining ring DIN 471 15 x 1.0 (groove 14.3 x 1.1)
14	1	SL-202	Flanged lead-screw nut TR16x4 RH, bronze CuSn7 (RG7): flange 48 x 10, body 28 x 30, 4 x 6.6 on 38 PCD (DIN 103 thread; envelope per typical catalogue nut, check vendor drawing)
15	4	SL-511	Socket head cap screw ISO 4762 M6 x 20, 8.8 (flange nut to carriage, 4)
16	1	SL-201	Lead screw TR16x4 right-hand, single start, rolled C45, 666 long: 595 thread, end machined to 10 h6 x 71 (51200 + 6200 seats, clamp coupling hub, no keyway)
17	1	SL-203	Drive block, C45 machined: fixed end of the screw, 51200 thrust bearing (screw thrust in compression onto the thread shoulder) + 6200-2RS radial; 4 x M8 to the base drive plate
18	1	SL-403	Thrust ball bearing 51200 10 x 26 x 11 (C 12.7 kN, C0 18.6 kN, catalogue - confirm); takes the screw thrust in compression
19	1	SL-402	Deep groove ball bearing 6200-2RS 10 x 30 x 9 (C 5.07 kN, C0 2.36 kN), drive block radial
20	1	SL-204	Jaw coupling Lovejoy L-095 (OD 54, 63.5 long), clamp hubs bored 10 mm (screw, no keyway) and 16 mm (gearmotor), NBR spider 21.9 Nm nominal (catalogue, confirm)
21	1	SL-205	Geared BLDC motor StepperOnline M82B200-24P-30S/G82-71.5S1: 24 V, 200 W, 71.5:1 planetary, rated 33 Nm at 42 rpm (listing). Envelope model: flange / shaft / lengths assumed, check the vendor drawing
22	4	SL-512	Socket head cap screw ISO 4762 M8 x 20, 8.8 (drive block to base, 4)
23	4	SL-513	Socket head cap screw ISO 4762 M6 x 16, 8.8 (gearmotor to motor plate, 4)

DRAWN Jay Patel	10/29/2026	
CHECKED		
QA		TITLE: ScissorLift - general assembly
MFG		
APPROVED		SIZE D | DWG NO SL-DWG-001 | REV
		SCALE 1 / 5 | SHEET 2 OF 2